

Can Ambient WiFi Replace LiDAR for Automotive SLAM?

Localization Yes, Mapping No — and Why

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Abstract—LiDAR dominates the sensor cost of an autonomous vehicle, which motivates a blunt question: can ambient WiFi sensing simply take its place? We answer it with the first head-to-head, on-vehicle, outdoor comparison of WiFi and LiDAR SLAM evaluated on the *same* ray-traced scenes, against the *same* ground truth, with the *same* six metrics. The study is simulation-based (Sionna RT), anchored by a real-LiDAR control: our SLAM back-end run on KITTI odometry attains 1.16 m aligned ATE over a 394 m drive ($\approx 0.3\%$ drift), which is consistent with real LiDAR odometry. Three findings follow. First, on *localization* commodity-CSI WiFi is not merely competitive but superior: it reaches 0.027 m ATE, better than either of our LiDAR models, at 84–600 $\times$ lower sensor cost. Second, on *mapping* it fails outright (occupancy IoU ≈ 0 versus up to 1.00 for LiDAR), and no price buys coverage it cannot produce. Third, we locate the mapping ceiling precisely: it is *not* path discrimination, as previously supposed, but a front-end limit — roughly 89% of MUSIC detections correspond to *no real propagation path*, and a 6.45 m median range *bias* (far beyond the 0.94 m resolution limit) corrupts the rest. Consequently a learned discriminator cannot repair the map, and we show that a heuristic, a random forest, and a neural network all fail identically. The practical consequence is a hybrid: adding WiFi to an existing LiDAR costs $\approx 0.5\%$ more and improves localization by 36–79% — provided the two sensors are of comparable accuracy, since a badly mismatched pair degrades the stronger one.

Index Terms—Internet of Things, SLAM, Wi-Fi, channel state information, LiDAR, sensor fusion, autonomous vehicles, cost analysis, ray tracing.

I. INTRODUCTION

LIDAR is the reference exteroceptive sensor for vehicle SLAM, and it is also the expensive one: a mid-range spinning unit of the class routinely used in research costs thousands of dollars, while the WiFi silicon already present in a vehicle costs tens. If ambient WiFi — signals that are radiated anyway, by infrastructure that already exists — could stand in for LiDAR, the sensing bill for autonomy would change by orders of magnitude. That is the question this paper asks, and it asks it without hedging: *can WiFi replace LiDAR, or not?*

The honest answer turns out to be neither “yes” nor “no” but a boundary, and the boundary is the contribution. WiFi replaces LiDAR for one half of SLAM and fails at the other, and the failure is not where the literature — including our own earlier work [?] — assumed it was.

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Code, data, and every figure-generating script are released openly; all reported numbers are reproduced from the committed artifacts.

A. Contributions

- 1) **The first on-vehicle, outdoor, head-to-head WiFi-vs-LiDAR SLAM comparison.** Both modalities are evaluated on identical ray-traced scenes against identical footprint ground truth with an identical metric suite, and the LiDAR back-end is externally validated on real KITTI data [?] ($\approx 0.3\%$ drift). Prior WiFi-SLAM is indoor [?], or uses WiFi to *assist* a primary ranging sensor [?], [?]; none tests replacement on a vehicle (Section II).
- 2) **A sensor-cost analysis** — to our knowledge the first for WiFi-vs-LiDAR SLAM — including *cost-normalized* accuracy and coverage. WiFi’s sensing package is **84–600 $\times$** cheaper than the LiDAR class we simulate, and delivers two to three orders of magnitude better accuracy per dollar (Section ??).
- 3) **A symmetric fusion study with a cost verdict.** Fusing the two helps *conditionally*, and we identify the condition (sensor parity): fusion beats both solo sensors in three of four configurations but *degrades* the stronger sensor by 8 $\times$ when the pair is badly mismatched (Section ??).
- 4) **The mapping ceiling, located.** An isolation experiment shows the WiFi mapping floor is dominated by *phantom* detections ($\approx 89\%$) and estimator *range bias*, not by path discrimination. This refines the interpretation of [?] and tells a learned method where it must act (Section ??).

B. Relationship to our prior work

This paper extends [?], which established ambient WiFi as a feasible *radar*-replacement sensing front-end and reported an oracle-sensing map at 0.25 m accuracy alongside a realistic-sensing map bounded near 5 m. Its empirical results are reused here, not re-derived. Section ?? refines one of its *interpretations*: that work attributed the realistic-mapping floor to path discrimination and argued the discrimination was learnable. The isolation experiment below shows discrimination is in fact the *smallest* of three mechanisms, and that a learned discriminator therefore cannot lift the map.

II. RELATED WORK

WiFi as a standalone SLAM sensor. P2SLAM [?] is the closest prior art: it extracts two-way bearings from commodity CSI and integrates them with odometry in a GraphSLAM

back-end, achieving 26.9 cm median tracking. It is, however, evaluated *indoors only* (25×30 m and 50×40 m, with fixed anchors) and is benchmarked against *visual* SLAM, not LiDAR. Radio-fingerprint SLAM [?] does run outdoors on a mobile platform and uses radio as the primary modality, but it relies on RSS fingerprinting from smartphones rather than CSI, runs on a slow research UGV rather than a road vehicle, and attains its best accuracy only when fused with LiDAR.

WiFi as an assistant to a primary sensor. A larger body of work uses WiFi to *augment* a ranging sensor rather than replace it. ViWiD [?] offloads loop closure from a visual back-end to WiFi; [?] supplies WiFi fingerprint loop closures to a LiDAR SLAM that still performs the mapping; [?] fuses WiFi RSSI, LiDAR and IMU in an EKF. In every case LiDAR or a camera remains the mapping sensor. None of these works reports a WiFi-vs-LiDAR SLAM accuracy comparison, and none treats WiFi as a candidate replacement.

Learned RF sensing. Deep models can recover geometry from RF: RF-Pose [?] estimates human pose through walls from WiFi-band reflections, [?] regresses dense body pose from commodity CSI, and [?] reconstructs 3-D point clouds from CSI with a transformer. Notably these methods learn *end-to-end from raw CSI*; none classifies estimated propagation paths. Section ?? explains why that distinction is decisive.

The open cell. On-vehicle, outdoor, commodity-CSI WiFi sensing evaluated head-to-head against LiDAR for SLAM — with a cost analysis — is, to our knowledge, unoccupied. This paper occupies it.

distance, directional map-accuracy (estimate→truth) and map-completeness (truth→estimate), and occupancy IoU for the map. Reducing LiDAR to 2-D is a deliberate, stated choice: it is what makes the comparison apples-to-apples, and it is discussed as a limitation in Section ??.

III. SYSTEM MODEL

Both modalities are driven through one pipeline and scored on one plane, so that any difference between them is attributable to the sensor and not to the estimator, the scene, or the metric (Fig. 1).

Scenes and ground truth. Two ray-traced outdoor scenes are used throughout: a *controlled wall* (a large metal reflector, giving clean single-bounce geometry) and a reflective *street canyon* with buildings and cars. Ground truth is the bird's-eye footprint of every non-floor scatterer, identical for both sensors.

WiFi path. Ambient access points illuminate the scene; the vehicle receives commodity CSI, from which a joint 2-D (delay–azimuth) MUSIC front-end estimates each path's bistatic delay and arrival azimuth. Each detection is triangulated on the bistatic ellipse — the reflector lies where the measured path length $|AP \rightarrow R| + |R \rightarrow \text{veh}|$ and the arrival bearing intersect — and the resulting reflectors, together with the same detections used as particle weights, drive a Rao-Blackwellized particle-filter SLAM.

LiDAR path. The LiDAR is reduced to the same plane: a horizontal ring produces a 2-D scan, which a scan-to-map ICP registers against an accumulated voxel map. Because a single LiDAR model would beg the question of *which* LiDAR, we bracket the sensor with two models (Section ??).

The common comparison plane. Trajectories are *xy*, maps are 2-D point sets, and both sensors are scored with the same six metrics: ATE and RPE for the trajectory; Chamfer

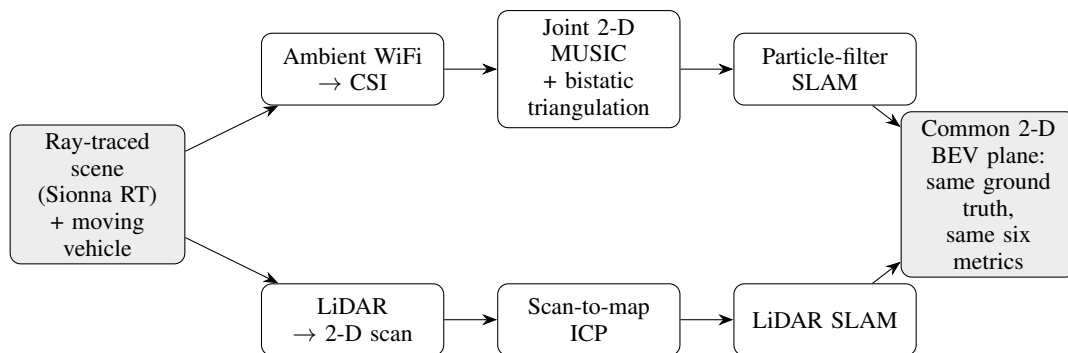


Fig. 1. Both sensors are driven through the same scene and scored on the same 2-D bird's-eye plane, against the same footprint ground truth, with the same six metrics. Any difference is then attributable to the sensor, not to the estimator or the metric.